

RIDOT Symposium

**Artificial Intelligence in the
Road & Mobility Sector**

Operational AI Solutions
& AI Pilot Projects

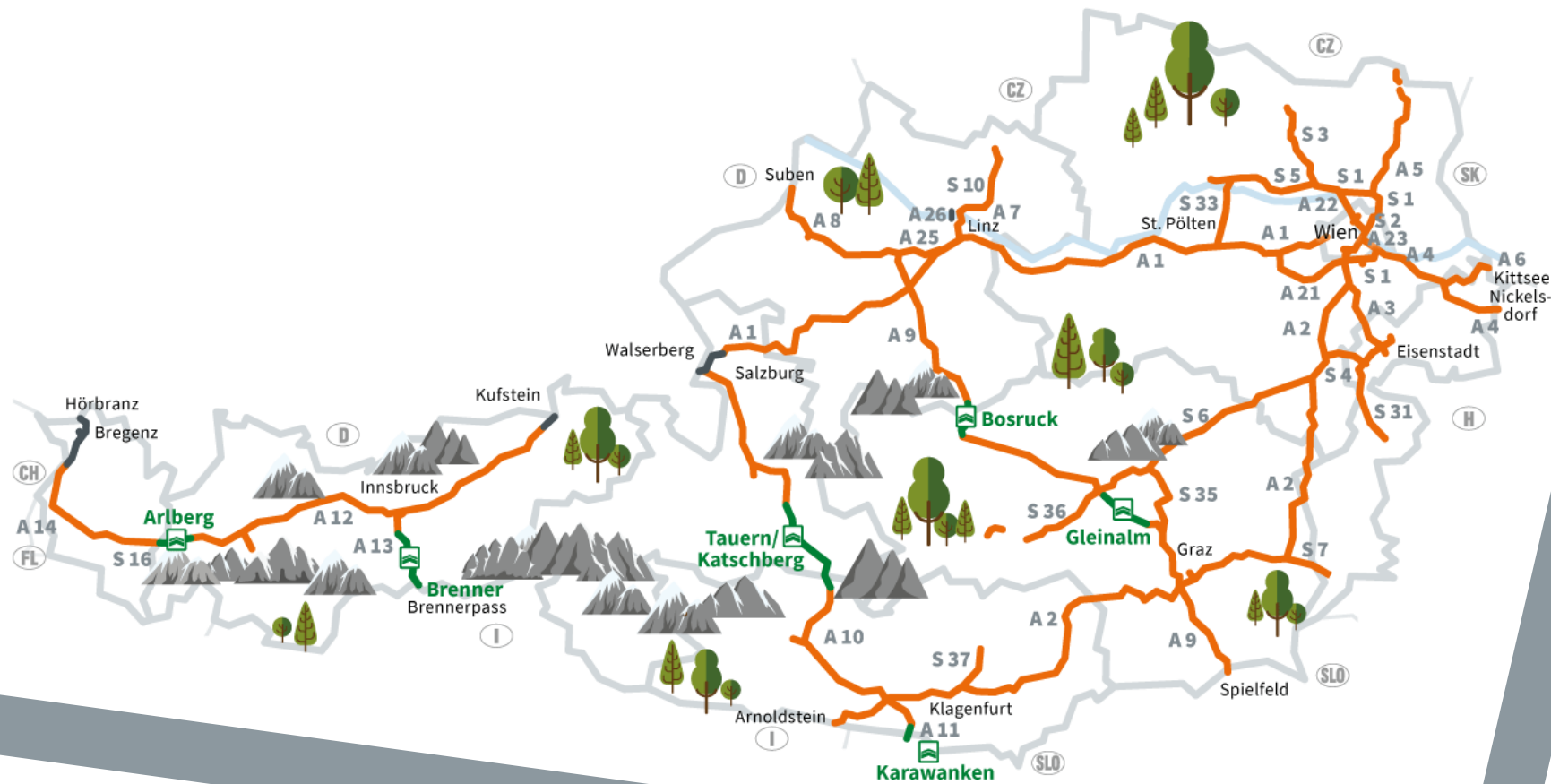


A|S|F|i|N|A|G

GUTE FAHRT, ÖSTERREICH!

ASFINAG (AUSTRIA)





~3,500
Employees

6
Special toll stations
(digital products)

9
Traffic management
centres

33.05 BILLION

km of total traffic
per year

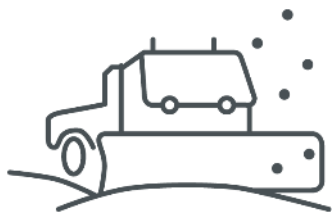
2,266

kilometres of
road network

43

Motorway operation
and maintenance
facilities

SCOPE OF ACTIVITIES OF ASFINAG



OPERATION



MAINTENANCE



CONSTRUCTION

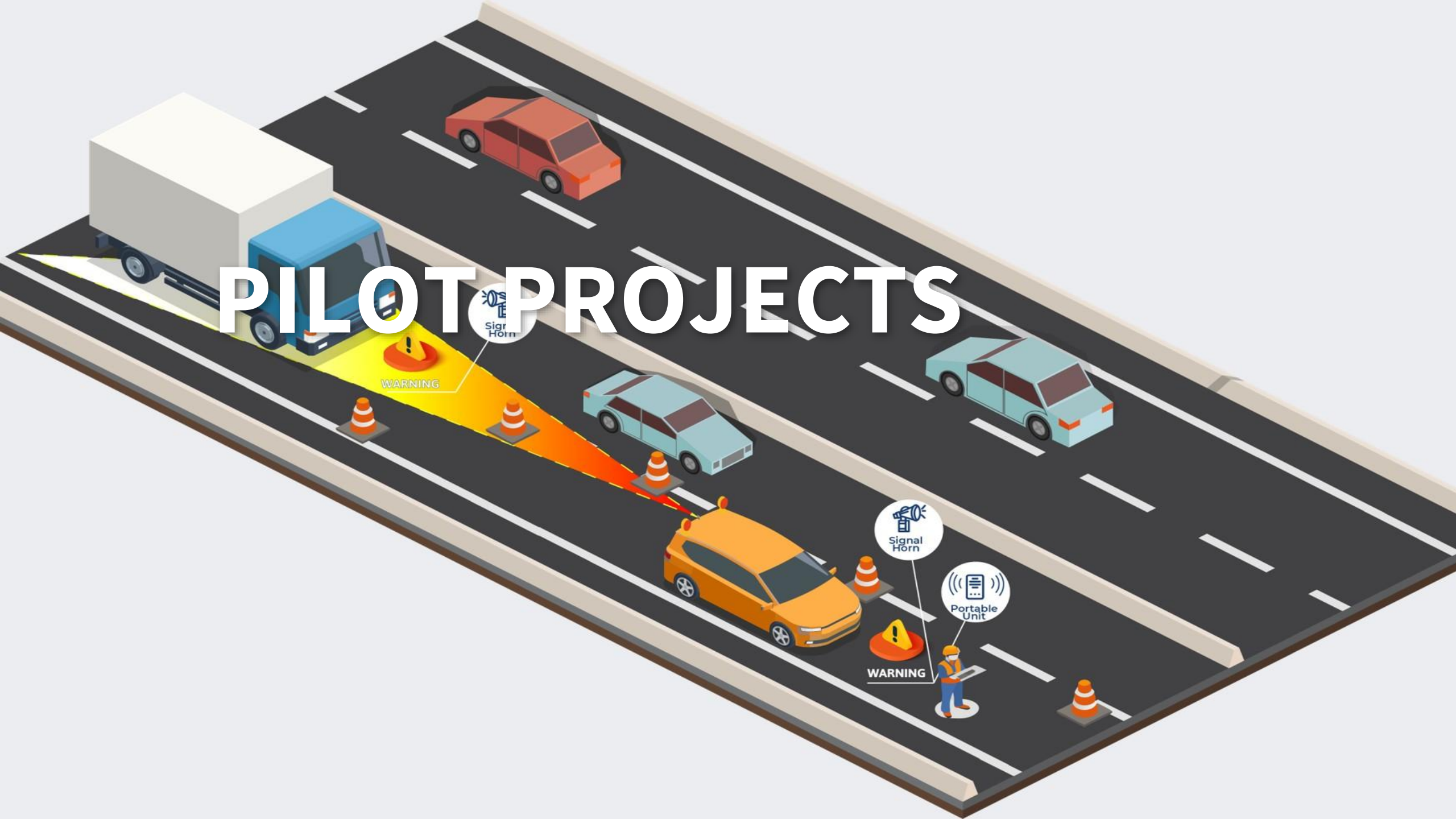


TOLL COLLECTION



FINANCING

PILOT PROJECTS

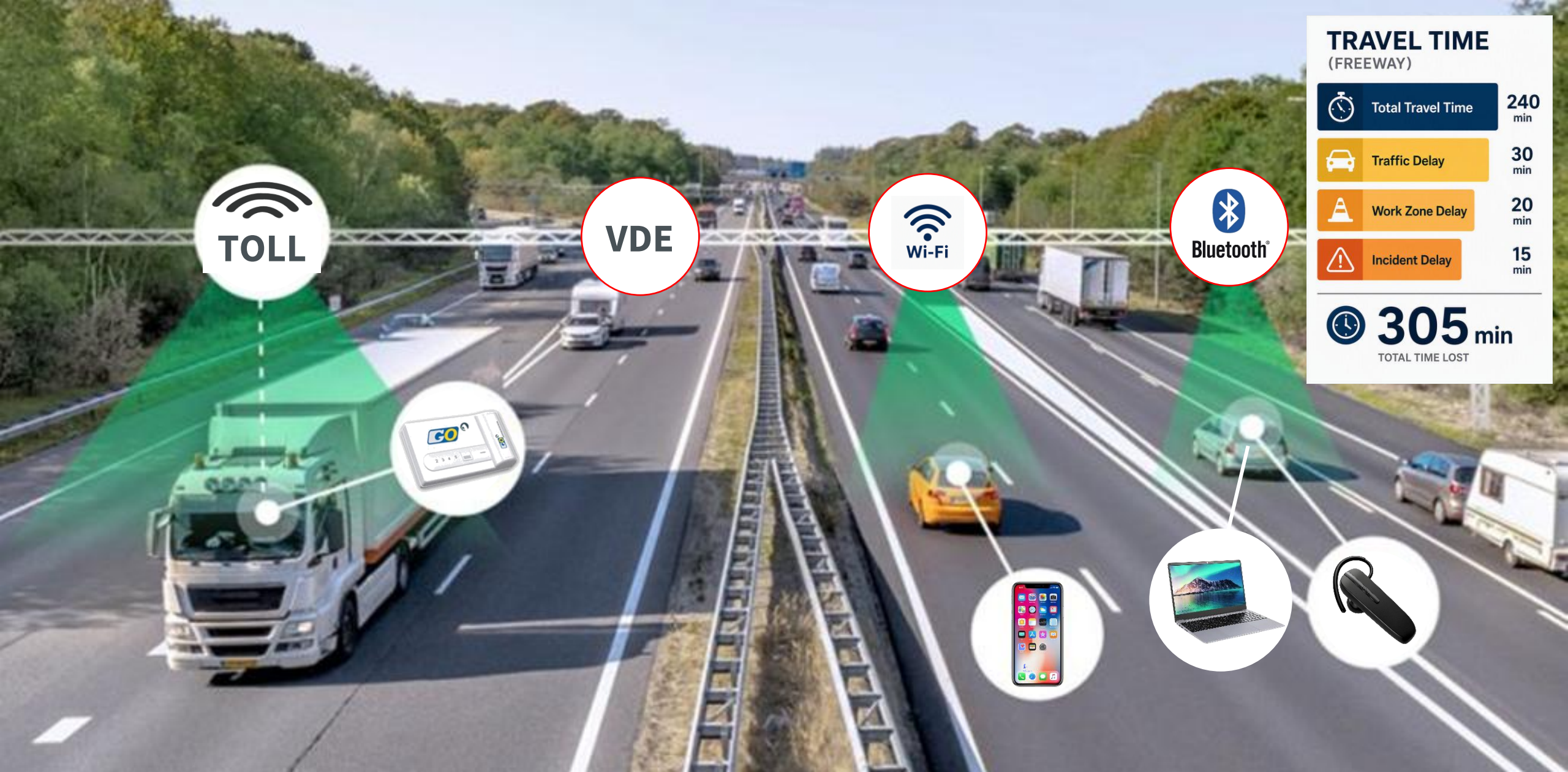


ROAD SAFETY / WORKER SAFETY

COLLISIONEYE® VIDEO



REAL-TIME TRAVEL TIME MEASUREMENT



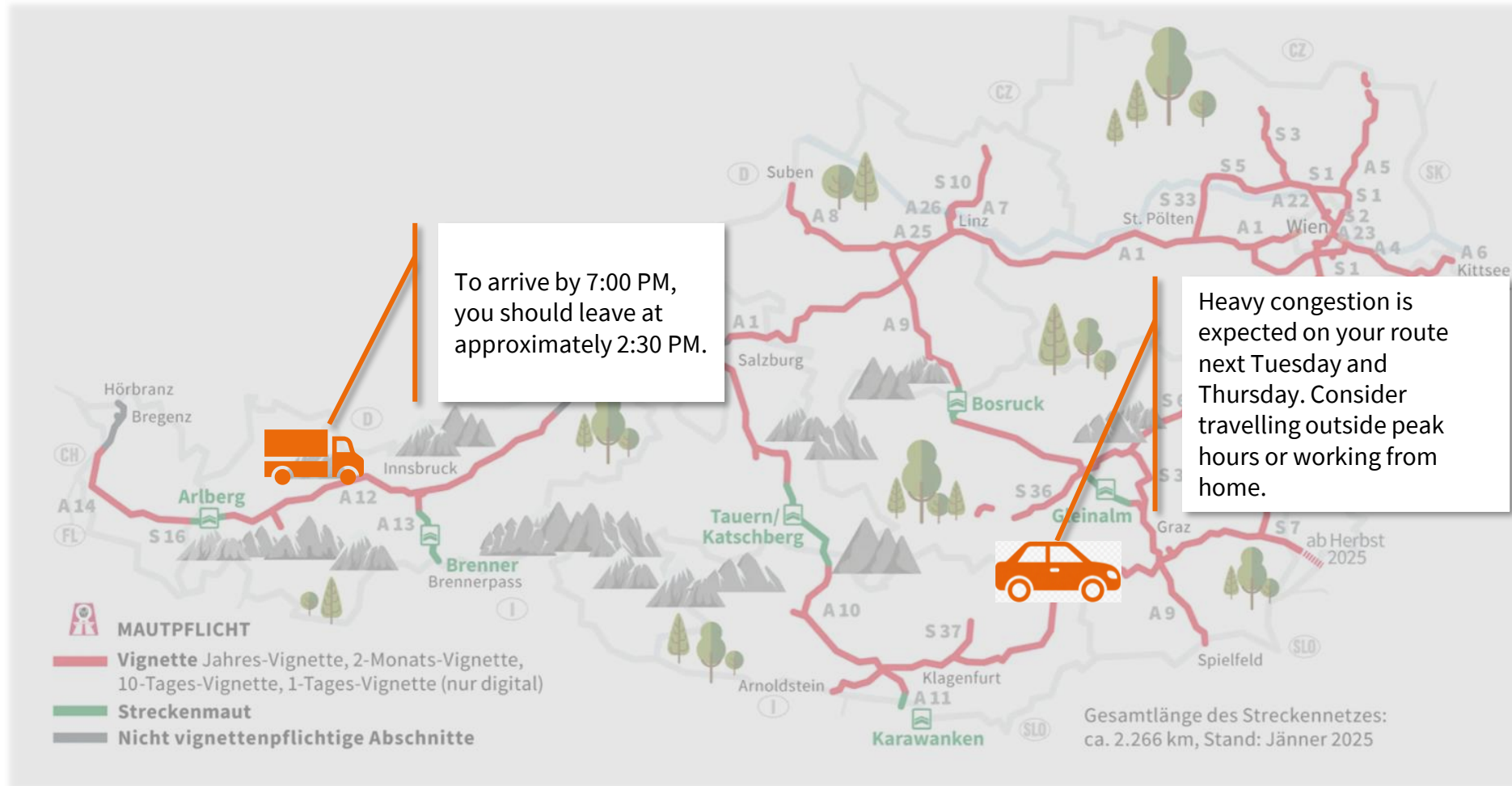
TRAVEL TIME (FREEWAY)

	Total Travel Time	240 min
	Traffic Delay	30 min
	Work Zone Delay	20 min
	Incident Delay	15 min

 **305 min**
TOTAL TIME LOST

TRAVEL TIME PREDICTION

FROM TRAFFIC DATA TO CUSTOMER-ORIENTED PREDICTIONS



OVERVIEW GEN AI @ ASFINAG



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DEVELOPMENT OF FILE & INFORMATION SEARCH

Local File Search

Phase 1
(1990s – now)

File names,
on-demand search

I-Drive

Centralized enterprise search

Phase 2
(2000s – now)

Central indexes,
pre-built index, full-
text and metadata
search

SharePoint

Modern vector search and LLMs

Phase 3
(since 2022)



PRIVACY FIRST: NO DATA LEAVES OUR ENVIRONMENT



General Data Protection Regulation GDPR background Concept. Free Vector / www.vecteezy.com

GDPR (European Union Framework, 2018)

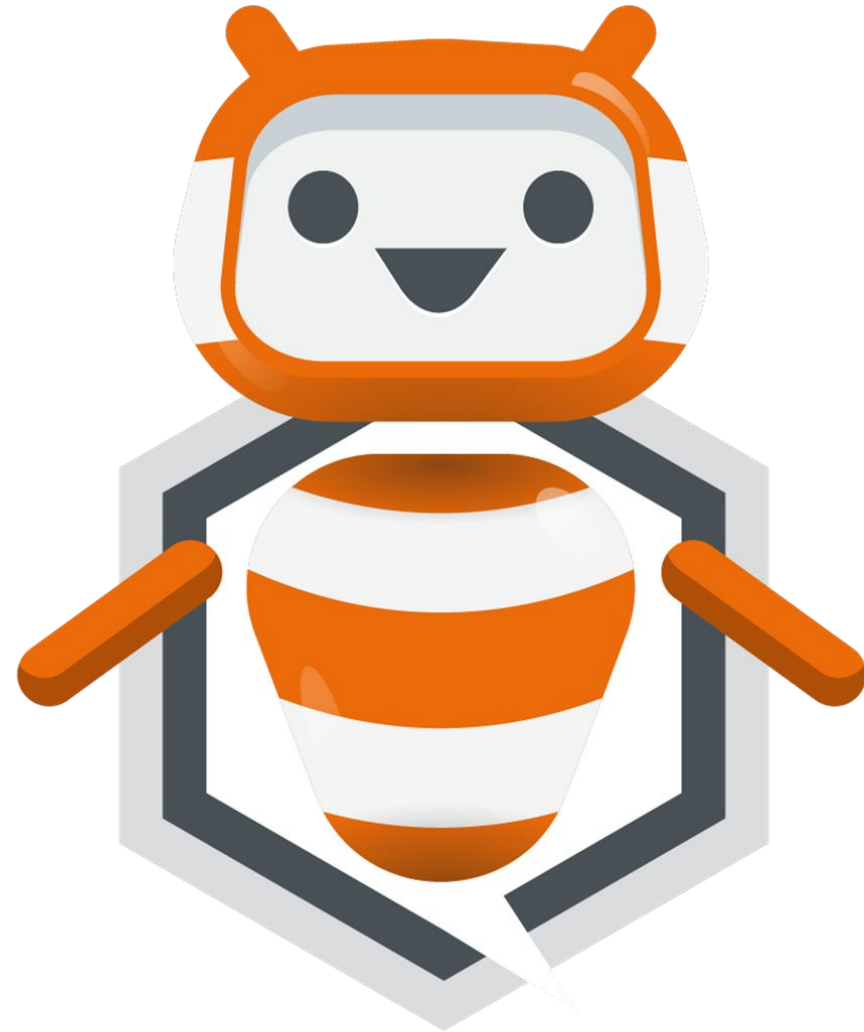
- ✓ adequate level of data protection
- ✓ No vendor training on your data
- ✓ Right to erasure & access
- ✓ Mandatory data breach notification
- ✓ Heavy fines for violations (4% global revenue)

AGNES



AND

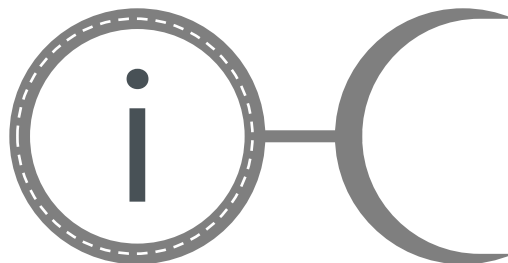
ADAM



INTRODUCING AGNES



A|S|F|i|N|A|G



Employees can ask questions in natural language and receive answers based on internal ASFINAG knowledge.



Behind the scenes, AGNES uses OpenAI models, but all information remains within ASFINAG-controlled systems.



One design principle was particularly important: Employees must be able to trust the system.

Meet Maria.

New ASFINAG employee.
First month on the job.

*"I need to submit an expense report.
Where do I find the form?
Which rules apply to that at ASFINAG?
Who approves it?"*

Before AGNES, Maria had to search:

The I-Drive (maybe)?

SharePoint (1.400 documents)?

Ask a colleague (also unsure)?

Email attachment (from 2022, you don't know)?

 *Average time lost: 30 min per employee, per question*

With AGNES Maria types:

"How do I submit an expense report?":

Here is the expense process:

1. Download Form [LF-009](#)
2. See Guideline [RL-Finance § 4.2](#)
3. Submit via [ADONIS workflow](#)

 *Source: RL-Finance § 4.2 | LF-009 (Expense Form)*

AGNES: 2 YEARS OF LIFE IN NUMBERS



more than 1,700 registered users



ø 600 active users per month

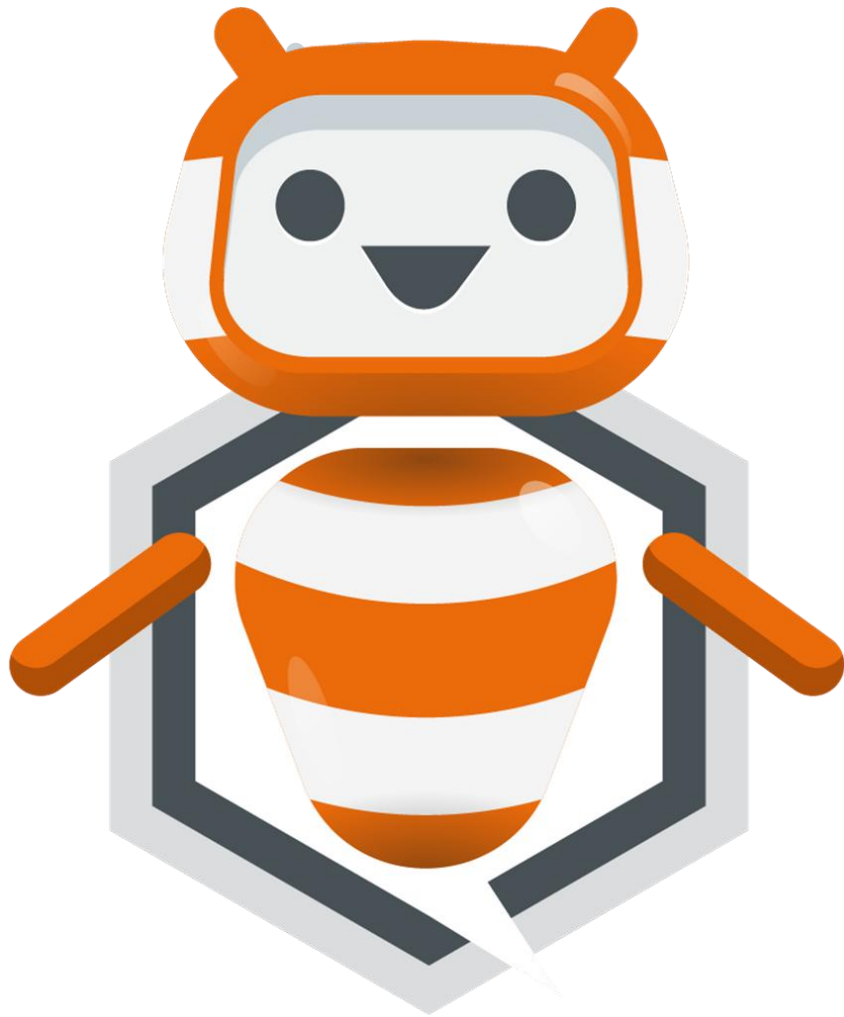


answered more than 50,000 questions

2014 x thumbs up, 963 x thumbs down



INTRODUCING ADAM



Meet Kathi.

Kathi is a noise protection specialist.

The noise from the highway is unbearable. We are waiting for the noise barrier ASFINAG promised 5 years ago. Nothing has happened.

Before ADAM, Kathi often had to search through years of correspondence:

SharePoint Request System

The I-Drive (scanned paper letter)?

*Ask colleagues from other departments
(who knows this person and promised it?)*



Average time lost: several hours per inquiry

After Kathi set up a personal chatbot, she copies the request into ADAM.

ADAM identifies a previous case from the same address and retrieves the response from five years ago:

“Dear Mr. Green,
we thank you for your inquiry and can respond to you as follows ...

THE DIFFERENCE IS ...



VS.



Enterprise-wide chatbot

Content ninjas are responsible for the content
and keep it up to date

Personal chatbot (for teams / groups)

Content can be selected by yourself
Teams / people are responsible for the content
themselves

"Think of AGNES as the company librarian."

"Think of ADAM as a specialist working next to you."

OUR TRAINING STRATEGY



AI Fundamentals

How LLMs work (non-technical)

What AI can & cannot do

Prompt engineering basics

AI ethics & bias awareness



Practical Toolbox

AGNES: how to ask good questions

ADAM: building your own bot

Document analysis workflows

Privacy rules in practice



Advanced & Specialist

Prompt design for specific domains

Evaluating AI output quality

Bot creation & knowledge base mgmt

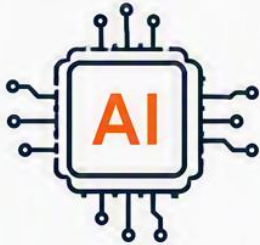
AI governance for managers

Training is meanwhile delivered primarily as self-paced microlearning — accessible to all 3,500 employees at any time.

Technology is only **20%** of the challenge —
people and culture are **80%**

TECHNOLOGY

20%



PEOPLE & CULTURE

80%



Give AI a
Face & Name



guarantee
privacy



Invest in
communication



Measure Quality
Always

Image source: Copilot (GPT 5.5)

THANK YOU

